Jian Feng

j36feng@uwaterloo.ca | 647-766-0178 | Website | linkedin/in/fenjian | github/faketut

EMPLOYMENT

Manulife Financial Corporation

Co-op 2026 –

- Cut AKS namespace onboarding from days to minutes by automating ServiceNow tasks into Terraform/GitOps PRs with auto change tickets.
- Built a Foundry LLM agent that validates cert-SAN gitops PRs against runbooks across 9 AKS clusters, replacing manual review and catching misconfigs before prod.
- Built Power Apps-driven DL automation framework reducing manual membership updates to minutes via event-driven Graph API sync.

Treasury Board of Canada Secretariat

Co-op 2025

- Shipped a .NET MVC module auto-normalizing job titles across a 5K-firm public dataset, unblocking analyst peer-group and governance reporting.
- Migrated 30 Selenium cases (5 suites) to Katalon on Azure DevOps CI/CD; demoed to execs and adopted by other QA teams to ease no-code onboarding.
- Refactored core modules of the enterprise data platform and tuned SQL, cutting 5K-firm-facing browser page load 30% for faster reviews.

Lansi Institute of Artificial Intelligence in Medicine

Co-op (Remote) 2024

- Built genomic ETL pipelines for ~100GB multi-omics data; per-sample fan-out via Python `multiprocessing` cut runtime 89%.
- Ran XGBoost + SMR/MR² across a 68-study T2D meta-analysis as research assistant; narrowed the candidate drug-target shortlist used in the published paper.

Sanyang Technology Co., Ltd.

Intern 2024

- Built Gantt visualizer of a flexible job-shop scheduling system, unblocking algorithm A/B runs on benchmark instances.
- Shipped Spring Boot CRUD APIs for cloud reporting/BI modules, unblocking front-end dashboards and downstream ETL consumers.

EDUCATION

University of Waterloo 2023-2028

Bachelor of Computing and Financial Management

- Majors: Computer Science, Finance, Statistics
- Scaled UX Research Hub 32× (10→318 members) as Director; negotiated Google UXR keynote at $0 cost and brokered TikTok chapter cross-promotion for sustained recruitment growth.
- Velocity Co-op Problem Award (top submission among co-op cohort); 3rd Place, SAF Investment Research Challenge; Finalist, SAF Strategy Case Competition
- Certifications: Duke Data Engineering, IBM Vector Database, UCI Database Design & Operational BI, Google Data Analytics, Google IT Support

PERSONAL PROJECTS

CabStream ETL

- Terraform-provisioned GCP stack (GCS + BigQuery + Airflow); whole pipeline tears down and rebuilds from one `apply`.
- dbt three-layer model (staging → dims/facts → marts) with `schema.yml` freshness/uniqueness tests; Airflow DAG logic unit-tested in CI.
- External-table-then-CTAS pattern keeps raw parquet cheap on GCS while marts stay query-optimised in BigQuery.

MiniKV

- Go LSM engine — CRC32C WAL, skip-list MemTable, snappy SSTables with sparse index + file-level Bloom so negative lookups skip disk.
- Serializable OCC: snapshot reads, buffered writes, commit-time read-set validation vs a bounded commit log — atomic on no conflict.
- Async leader/follower replication over HTTP NDJSON and gRPC; followers resume from local `LastSeq()` on reconnect, no event replay.

ArmyOfSafeguards

- Four DeBERTa safeguard heads (toxicity, sexual, jailbreak, factuality) run in parallel via `expert_runner`, hot-swappable via env vars.
- Learned logistic meta-aggregator fuses per-expert probabilities into a calibrated verdict, beating threshold baselines on held-out HF.
- Training pipeline turns HF labels into expert-Q features and meta JSONL, so adding a new expert needs no aggregator changes.

Anchor

- Splunk drift agent fingerprints a healthy window, then ranks diffs against later windows with LLM-narrated SPL suggestions.
- Feedback loop reweights signals and surfaces recurring unresolved drifts as blind spots.

SKILLS

Python, SQL, C#, PostgreSQL, MySQL, MongoDB, Redis, AWS, GCP, Azure, Docker, Kubernetes, Airflow, dbt, Power BI, Tableau, Excel,